

Vector Spaces and Linear Maps

The matrix is not the point. The matrix is a shadow the point casts once you pick coordinates.

WHERE WE ARE

Linear algebra is not the study of matrices. It is the study of *things that add and scale* — and a matrix is what one of those things looks like after you have made an arbitrary choice of how to describe it. Almost every difficulty people have later in this book traces back to having learned the shadow instead of the object.

Physics is overwhelmingly linear, and for two reasons that are worth keeping strictly apart. The first: **we linearize everything**. Chapter 0.1 replaced functions by their best linear approximations, and Chapter 0.3 made that into a controlled expansion. When the linear term is the first surviving one, linear algebra is the mathematics of the leading behaviour of anything at all. The second reason is different in kind: **in quantum mechanics superposition is not an approximation**. If a system can be in state $|\psi_1\rangle$ and in state $|\psi_2\rangle$, it can be in $\alpha|\psi_1\rangle + \beta|\psi_2\rangle$, exactly, with no small parameter and no correction terms. Quantum mechanics is not *approximately* linear algebra; from Chapter 4.2 onward it simply *is* linear algebra, and the physical content lives in which vector space and which operators.

So this chapter builds the object first and the array second. By the end you will know why matrix multiplication has the shape it has (it is forced — it is not a convention), what a determinant is before anyone writes down a cofactor, and why the trace — an apparently trivial sum of diagonal entries — turns out to be one of the most reused quantities in theoretical physics.

TOOLS YOU'LL NEED — Chapter 0.1: linearization, and the characterisation of the exponential as the function equal to its own derivative (we use it once, in §6). Nothing else. Everything about vectors is built here from scratch.

1 · What a vector actually is

You were probably told a vector is an arrow: a thing with magnitude and direction. That description is a special case of a special case, and it will strand you in Chapter 4.2, where the vectors are functions and there is not an arrow in sight.

Here is the actual definition. A **vector space** over a field of scalars $\mathbb{F}$ (for us always $\mathbb{R}$ or $\mathbb{C}$) is a set V equipped with two operations — addition of two elements of V , and multiplication of an element of V by a scalar — whose results again lie in V , and which satisfy eight rules:

$$\begin{array}{ll}
(\text{A1}) & u + v = v + u \\
(\text{A2}) & (u + v) + w = u + (v + w) \\
(\text{A3}) & \exists 0 : v + 0 = v \\
(\text{A4}) & \exists (-v) : v + (-v) = 0 \\
(\text{S1}) & 1v = v \\
(\text{S2}) & a(bv) = (ab)v \\
(\text{S3}) & a(u + v) = au + av \\
(\text{S4}) & (a + b)v = av + bv
\end{array} \tag{0.4.1}$$

That is the whole definition. Notice what is *not* in it: no length, no angle, no coordinates, no arrows. Those come later and cost extra — length and angle are Chapter 0.5, coordinates are §2 of this chapter and are always a choice.

The axioms are not decoration; they do work. Two examples, both of which you have used a thousand times without noticing they need proving.

The scalar 0 annihilates. Nothing in (0.4.1) says $0v$ is the zero vector — the symbol 0 on the left is a number and on the right a vector, and they are different objects. But by (S4), $0v = (0 + 0)v = 0v + 0v$. Add $-(0v)$ to both sides and use (A4), (A3): $0 = 0v$. Similarly $(-1)v = -v$, because $v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0$, and by (A4) the thing you add to v to get 0 is what " $-v$ " means.

The examples that are not arrows

1. Tuples, $\mathbb{R}^n$. Add componentwise, scale componentwise. This is the familiar one, and it is the *only* one that looks like arrows.

2. Polynomials of degree at most n . Call this space P_n . The sum of two polynomials of degree $\leq n$ has degree $\leq n$; a scalar multiple likewise; the zero polynomial is the zero vector. All eight axioms hold because they hold coefficient by coefficient. Note the trap: polynomials of degree *exactly* n do *not* form a vector space — x^2 and $-x^2 + x$ are both of degree exactly 2, and their sum is not, and the zero polynomial is missing entirely. Closure is a real condition.

3. The solutions of a linear differential equation. This one matters more than the others, so we will do it properly. Consider

$$L[y] \equiv y'' + p(x)y' + q(x)y = 0. \tag{0.4.2}$$

Suppose y_1 and y_2 both solve it. Then for any scalars a, b , using only the fact that differentiation is itself additive and commutes with constants,

$$\begin{aligned}
L[ay_1 + by_2] &= a(y_1'' + py_1' + qy_1) + b(y_2'' + py_2' + qy_2) \\
&= a \cdot 0 + b \cdot 0 = 0.
\end{aligned} \tag{0.4.3}$$

So the solution set is closed under addition and scaling: **it is a vector space**. That statement has a name in physics — *superposition* — and it is the reason you may add two solutions of the wave equation and get a solution, the reason normal modes work (Chapter 0.8), and the reason a plucked string can be decomposed into harmonics (Chapter 0.9). It is not a physical principle. It is the observation that L is a linear map and that the set of things a linear map sends to zero is a vector space.

The converse trap is worth seeing now, since it explains a rule you were once made to memorise. If the equation is *inhomogeneous*, $L[y] = f$ with $f \neq 0$, the solution set is not a vector space: the sum of two solutions gives $2f$, not f . But if y_p is any one solution and y is any other, then $L[y - y_p] = f - f = 0$, so the difference solves the homogeneous equation. Hence *general solution* = *particular solution* + (*vector space of homogeneous solutions*). That is a plane in a space, shifted off the origin; the "arbitrary constants" you were told to add are coordinates on the shifted plane.

4. Functions on an interval. All real-valued functions on $[a, b]$, added pointwise: $(f + g)(x) \equiv f(x) + g(x)$, $(cf)(x) \equiv c f(x)$. Every axiom holds pointwise because it holds for real numbers. This space is *infinite-dimensional* in the sense §2 is about to make precise, and it is the arena of Chapters 0.9 and 4.3.

5. Matrices. The $m \times n$ real matrices form a vector space under entrywise addition. Read that twice: *matrices are themselves vectors*. If you think "vector = column of numbers" you cannot even parse the sentence, and you will need it in Chapter 6.1, where the vector space in question is a space of matrices and its elements are the generators of a symmetry group.

6. Quantum states. The states of a quantum system form a vector space over $\mathbb{C}$. Not metaphorically — the superposition principle is precisely the assertion that the axioms (0.4.1) hold for physical states. This is why Chapter 4.2 is a linear algebra chapter with physics vocabulary sprinkled on top.

THE DEFINITION TO CARRY FORWARD

"Vector" does not mean "arrow". It means "element of a vector space". The word is a statement about what you are allowed to do to the object — add it to another one, scale it — and about nothing else.

A function, a polynomial, a matrix, a solution of a differential equation, and a quantum state are all vectors, and every theorem in this chapter applies to all of them without modification. That is not generality for its own sake: it is why the same eigenvalue machinery solves a coupled-oscillator problem in Chapter 0.8 and a hydrogen atom in Chapter 4.5.

2 · Span, independence, basis, dimension

We now build the machinery that lets us describe an abstract vector by a list of numbers — and, just as importantly, that tells us the list is not the vector.

A **linear combination** of $v_1, \dots, v_k$ is any vector $c_1 v_1 + \dots + c_k v_k$. The **span** of a set is the collection of all its linear combinations; it is itself a vector space (a sum of two linear combinations is a linear combination, and so is a multiple of one). The set $\{v_1, \dots, v_k\}$ is **linearly independent** if

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k = 0 \implies c_1 = c_2 = \dots = c_k = 0, \quad (0.4.4)$$

i.e. the only way to build the zero vector out of them is to use none of them. A **basis** is a set that is both linearly independent and spans the whole space.

Coordinates exist, and they are unique

Let $\{e_1, \dots, e_n\}$ be a basis of V . Because it spans, every $v \in V$ can be written as $v = \sum_j x_j e_j$ for some scalars x_j . The claim is that those scalars are *uniquely determined* by v . Suppose two expansions:

$$v = \sum_{j=1}^n x_j e_j = \sum_{j=1}^n y_j e_j \implies \sum_{j=1}^n (x_j - y_j) e_j = 0 \implies x_j = y_j \quad \forall j, \quad (0.4.5)$$

the last step being exactly the definition of independence (0.4.4) applied to the coefficients $c_j = x_j - y_j$. The converse is just as quick: if every vector has a unique expansion, apply that to $v = 0$, which certainly has the expansion with all coefficients zero; uniqueness says there is no other, which is independence. So:

BASIS, RESTATED

A set is a basis exactly when **every vector in the space has one and only one coordinate list with respect to it**. Spanning gives you *at least* one; independence gives you *at most* one.

The scalars x_j are the **coordinates** of v in the basis $\{e_j\}$, and we will write them as a column and give the column its own symbol, deliberately different from the vector's:

$$v = \sum_{j=1}^n x_j e_j \quad \longleftrightarrow \quad \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \quad (0.4.6)$$

Throughout this chapter, v is the vector and $\mathbf{x}$ is its coordinate list. They are not the same kind of thing, and keeping them typographically apart is not pedantry — it is the entire subject.

Dimension

The **dimension** of V is the number of elements in a basis. For that to be a definition rather than a hope, we need to know that two different bases cannot have different sizes. They cannot, and the proof is a swapping argument worth seeing once (grind box below). Given it, the examples fall out:

SPACE	A BASIS	DIMENSION
$\mathbb{R}^n$	$e_1, \dots, e_n$ (the standard columns)	n
P_3 , polynomials of degree ≤ 3	$1, x, x^2, x^3$	4
2×2 real matrices	the four matrices with a single 1	4
Solutions of $y'' + \omega^2 y = 0$	$\cos \omega t, \sin \omega t$	2
Functions on $[a, b]$	— no finite basis exists —	∞

The polynomial entry deserves a word, since it is the one people wave through. Why are $1, x, x^2, x^3$ independent? Because if $c_0 + c_1x + c_2x^2 + c_3x^3 = 0$ as a function — true for every x — then the polynomial has infinitely many roots. A nonzero polynomial of degree ≤ 3 has at most three. So it is the zero polynomial and all $c_k = 0$. The independence of $\cos$ and $\sin$ is Problem 1, and it is proved the same way: evaluate somewhere convenient.

▼ Grind box — why every basis has the same size (the exchange lemma)

Lemma. If $\{w_1, \dots, w_m\}$ spans V and $\{v_1, \dots, v_k\}$ is independent, then $k \leq m$. An independent set can never be larger than a spanning set.

Proof. We swap the v 's in one at a time, keeping a spanning set at every step.

Since the w 's span, $v_1 = \sum_j c_j w_j$. Not every c_j is zero, since that would make $v_1 = 0$ and a set containing 0 is never independent (take $c_1 = 1$ and the rest zero in (0.4.4)). Relabel so that $c_1 \neq 0$; then

$$w_1 = \frac{1}{c_1} \left(v_1 - \sum_{j \geq 2} c_j w_j \right),$$

so anything expressible with w_1 is expressible with v_1 instead, and $\{v_1, w_2, \dots, w_m\}$ still spans V .

Now suppose we have got as far as $\{v_1, \dots, v_r, w_{r+1}, \dots, w_m\}$ spanning, with $r < k$. Expand the next one:

$$v_{r+1} = \sum_{i \leq r} a_i v_i + \sum_{j > r} b_j w_j.$$

If every b_j vanished, v_{r+1} would be a linear combination of $v_1, \dots, v_r$, and then $a_1 v_1 + \dots + a_r v_r - v_{r+1} = 0$ would be a nontrivial vanishing combination (the coefficient of v_{r+1} is $-1 \neq 0$), contradicting independence. So some $b_j \neq 0$; relabel it as b_{r+1} , solve for w_{r+1} as above, and swap it out. The new set still spans.

Each step consumes one w . If $k > m$ we would run out of w 's while v 's remained — but the step above proved that some b_j with $j > r$ is nonzero, which requires at least one w to still be present.

Contradiction. Hence $k \leq m$. ■

Consequence. Let B_1 and B_2 be bases with n_1 and n_2 elements. B_1 is independent and B_2 spans, so $n_1 \leq n_2$. B_2 is independent and B_1 spans, so $n_2 \leq n_1$. Therefore $n_1 = n_2$, and *dimension* is a property of the space, not of the description. ■

A corollary we need in §5. In an n -dimensional space, any n independent vectors already form a basis. For if they failed to span, there would be some u outside their span, and then $\{v_1, \dots, v_n, u\}$ would be independent — a vanishing combination $\sum c_i v_i + cu = 0$ must have $c = 0$ (otherwise $u =$

$-\frac{1}{c} \sum c_i v_i$ lies in the span), and then all $c_i = 0$ by independence of the v 's. That is $n + 1$ independent vectors in a space spanned by n , contradicting the lemma.

THE SENTENCE THAT GOVERNS THE REST OF THIS BOOK

The vector is the object. The coordinate list is a description of it relative to a choice you made. Change the choice and every number in the list changes, while nothing whatsoever happens to the vector.

This is the seed of two things that will otherwise look like arbitrary formalism when you meet them. In Chapter 2.4, a *tensor* is defined as an object whose components transform in the particular way that keeps the underlying object fixed while the description changes — the bookkeeping is §4 of this chapter, promoted to a definition. And in Chapter 3.2, the demand that *physics must not depend on the coordinates* becomes the organising principle of general relativity: a coordinate system is a choice of labels for events, no more physical than a choice of basis, so any equation whose content depends on that choice is not a law of nature. Both of those are this paragraph.

3 · Linear maps, and why matrix multiplication looks the way it does

A map $T : V \rightarrow W$ between vector spaces is **linear** if it respects both operations:

$$\begin{aligned} T(u + v) &= T(u) + T(v), & T(av) &= aT(v) \\ \iff T(au + bv) &= aT(u) + bT(v). \end{aligned} \tag{0.4.7}$$

Examples, deliberately not all of them matrices: rotation of the plane; differentiation $y \mapsto y'$ acting on polynomials or on smooth functions; the operator L of (0.4.2); definite integration $f \mapsto \int_a^b f$, which is linear from a function space to $\mathbb{R}$; and, in Chapter 4.2, every physical observable.

A linear map is determined by what it does to a basis

Let $\{e_1, \dots, e_n\}$ be a basis of V and let $T : V \rightarrow W$ be linear. Take any $v \in V$ and expand it in coordinates as in (0.4.6). Applying additivity $n - 1$ times and then homogeneity to each term,

$$T(v) = T\left(\sum_{j=1}^n x_j e_j\right) = \sum_{j=1}^n x_j T(e_j). \tag{0.4.8}$$

Read what that says: **if you know the n vectors $T(e_j)$, you know T completely** — its value on *every* one of the infinitely many vectors in V is then forced. And the converse holds too: choose any n vectors $w_1, \dots, w_n$ in W , whatever you like, and *define* $T(v) \equiv \sum_j x_j w_j$. This is a legitimate definition precisely because the coordinates x_j

are unique — that is what (0.4.5) was for — and it is linear because coordinates add and scale. So linear maps $V \rightarrow W$ correspond exactly to arbitrary lists of n vectors in W . Nothing is left over.

The matrix, defined

Now pick a basis $\{f_1, \dots, f_m\}$ of the target space W . Each $T(e_j)$ lives in W , so it has coordinates there, and we give them a name:

$$\boxed{T(e_j) = \sum_{i=1}^m T_{ij} f_i} \quad (0.4.9)$$

The $m \times n$ array of numbers T_{ij} is **the matrix of T in the bases $\{e_j\}, \{f_i\}$** . In words, and this is the sentence to remember: *the j -th column of the matrix is the coordinate list of the image of the j -th basis vector*. The first index labels which component in the target; the second labels which basis vector went in.

What does the array do to coordinates? Combine (0.4.8) with (0.4.9) and swap the order of two finite sums:

$$\begin{aligned} T(v) &= \sum_j x_j \sum_i T_{ij} f_i = \sum_i \left(\sum_j T_{ij} x_j \right) f_i \\ \implies (T(v))_i &= \sum_{j=1}^n T_{ij} x_j. \end{aligned} \quad (0.4.10)$$

That is the row-times-column rule for a matrix acting on a column vector. We did not postulate it. It is what (0.4.9) forces.

Composition, and the punchline

Here is the moment this section exists for. Let $T : U \rightarrow V$ and $S : V \rightarrow W$, with bases $\{u_k\}$ of U , $\{v_j\}$ of V , $\{w_i\}$ of W , and matrices defined by (0.4.9) in each case:

$$T(u_k) = \sum_j T_{jk} v_j, \quad S(v_j) = \sum_i S_{ij} w_i. \quad (0.4.11)$$

The composite map $S \circ T : U \rightarrow W$ — do T , then S — is linear (compose the definitions), so it has a matrix of its own, and by (0.4.9) that matrix is read off from what $S \circ T$ does to the basis $\{u_k\}$. So compute:

$$\begin{aligned} (S \circ T)(u_k) &= S(T(u_k)) = S\left(\sum_j T_{jk} v_j\right) = \sum_j T_{jk} S(v_j) \\ &= \sum_j T_{jk} \sum_i S_{ij} w_i = \sum_i \underbrace{\left(\sum_j S_{ij} T_{jk}\right)}_{\text{coefficient of } w_i} w_i. \end{aligned} \quad (0.4.12)$$

Compare the last line with the definition (0.4.9) and read off the matrix of the composite:

$$(ST)_{ik} = \sum_{j=1}^{\dim V} S_{ij} T_{jk} \quad (0.4.13)$$

There it is. **Matrix multiplication is not a convention someone chose; it is what composition of linear maps does to the arrays.** Every strange feature of the rule is now explained rather than memorised:

- **Why rows times columns.** The summed index j is the one that labels a basis of the *middle* space V — the space the output of T lands in and the input of S comes from. It must be summed because (0.4.12) feeds every component of $T(u_k)$ through S . The free indices i and k label the two ends.
- **Why the shapes must match.** An $m \times n$ times an $n \times p$ is legal and nothing else is, because the middle dimension is *the dimension of the middle space*. Mismatched shapes are not a typing error; they are an attempt to compose maps that do not connect.
- **Why ST means "first T ".** Because $(S \circ T)(u) = S(T(u))$ and the function nearest the argument acts first. The reversed-looking order is inherited from function notation, which is in turn inherited from writing $f(x)$ rather than $(x)f$.
- **Why it is associative for free.** $(RS)T = R(ST)$ requires no computation whatsoever: composition of functions is associative, since both sides mean "do T , then S , then R ". People verify this with triple sums. There is no need.
- **Why it need not commute.** Doing T then S is a different operation from doing S then T . See the warning box below, and then most of twentieth-century physics.

The repeated index j in (0.4.13), summed over while the free indices survive, is your first *contraction*. In Chapter 2.4 it gets a notation of its own — Einstein's convention, in which a repeated index is silently summed — and from there it never stops.

FAMILIAR GROUND

You already iterate a linear map professionally. Take a three-state disease model — progression-free (PF), progressed (P), dead (D) — and a cycle length of, say, three months. Let the state of a cohort be a *column* of probabilities $\mathbf{p}$, and let the transition matrix act on the left, $\mathbf{p}^{(n+1)} = P \mathbf{p}^{(n)}$. With that convention $P_{ij} = \Pr(\text{next state } i \mid \text{current state } j)$, so the *columns* sum to one. (Much of the health-economics literature writes the state as a row vector and multiplies on the right, in which case the rows sum to one instead. Both are correct; mixing them silently is the standard way to get a wrong answer.) With per-cycle probabilities 0.15 for progression, 0.05 for death from PF and 0.30 for death from P:

$$P = \begin{pmatrix} 0.80 & 0 & 0 \\ 0.15 & 0.70 & 0 \\ 0.05 & 0.30 & 1 \end{pmatrix}, \quad \mathbf{p}^{(0)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

n cycles is P^n — the matrix of the n -fold composite, by (0.4.13) applied $n - 1$ times. And the entries of P^2 are exactly the derivation in (0.4.12) read aloud: the chance of being progressed after two cycles is $(P^2)_{21} = \sum_j P_{2j} P_{j1} = 0.15 \times 0.70 + 0.80 \times 0.15 = 0.225$, a sum over the *intermediate* state j — progress-then-survive plus survive-then-progress. The summed middle index of matrix multiplication is a sum over all the ways of getting there.

The column-sum constraint has an algebraic reading. Writing $\mathbf{1}^\top$ for the row of ones, "every column sums to one" is $\mathbf{1}^\top P = \mathbf{1}^\top$: that row vector is left unchanged by P . A vector the map leaves alone is what §7 will call an *eigenvector*, here with eigenvalue 1, and this one is the algebraic form of "probability is conserved" — the total is the thing this map cannot change.

Everything else about the model's long run is likewise eigenvalues. The eigenvalues are the numbers λ for which $P - \lambda I$ fails to be invertible (§7 proves that is the same condition), and this P is triangular, for which §5 shows the determinant is the product of the diagonal. So $\det(P - \lambda I) = (0.8 - \lambda)(0.7 - \lambda)(1 - \lambda)$ and the eigenvalues are 1, 0.8, 0.7. The $\lambda = 1$ direction is the absorbing state — everyone is eventually dead — and the *rate* of approach is governed by the largest of the others: the PF fraction is exactly 0.8^n , and at $n = 10$ the cohort is 10.7% PF, 11.9% progressed, 77.4% dead. Which sub-unit eigenvalue dominates, and how fast, is the whole content of Chapter 0.5.

4 • Change of basis

Two people describe the same space with different bases. What is the relationship between their arrays? This is the most mechanical section of the chapter and the one with the longest reach: it is the prototype of every transformation law in relativity.

Let $\{e_j\}$ and $\{e'_j\}$ both be bases of V . Each new basis vector, being a vector, has coordinates in the old basis; call them P_{ij} :

$$e'_j = \sum_{i=1}^n P_{ij} e_i. \quad (0.4.14)$$

In words: *the columns of P are the new basis vectors, written in old coordinates*. P is invertible, and we can see it without appealing to anything: since $\{e'_k\}$ is also a basis, we may equally expand $e_i = \sum_k Q_{ki} e'_k$. Substitute one into the other:

$$e'_j = \sum_i P_{ij} \sum_k Q_{ki} e'_k = \sum_k \left(\sum_i Q_{ki} P_{ij} \right) e'_k \implies \sum_i Q_{ki} P_{ij} = \delta_{kj}, \quad (0.4.15)$$

where the implication is uniqueness of coordinates (0.4.5) again (the left side must be the expansion of e'_j in the basis $\{e'_k\}$, which is 1 in slot j and 0 elsewhere — that is what the Kronecker delta δ_{kj} means). So $QP = I$, and the same argument with the roles exchanged gives $PQ = I$. Hence $Q = P^{-1}$.

Components go the other way

Now take a vector v and expand it both ways, $v = \sum_j x_j e_j = \sum_j x'_j e'_j$. Substituting (0.4.14) into the second expansion,

$$v = \sum_j x'_j \sum_i P_{ij} e_i = \sum_i \left(\sum_j P_{ij} x'_j \right) e_i \implies \mathbf{x} = P \mathbf{x}', \quad \mathbf{x}' = P^{-1} \mathbf{x}. \quad (0.4.16)$$

Stare at that for a second, because it is the source of a lifetime of index confusion. The *basis vectors* transform with P ; the *components* transform with P^{-1} . They must go in opposite directions, because their product — the actual vector — does not change at all. Make your ruler twice as long and every measurement in rulers is halved. In Chapter 2.4 this opposition is given names, *covariant* and *contravariant*, and it is why some indices are written upstairs and some downstairs.

The similarity transformation

Finally, let $T : V \rightarrow V$ be a linear map from the space to itself — an *operator* — with matrix A in the old basis and A' in the new one:

$$T(e_j) = \sum_i A_{ij} e_i, \quad T(e'_j) = \sum_i A'_{ij} e'_i. \quad (0.4.17)$$

Compute $T(e'_j)$ two ways. First, push T through (0.4.14):

$$T(e'_j) = T\left(\sum_k P_{kj} e_k\right) = \sum_k P_{kj} T(e_k) = \sum_k P_{kj} \sum_i A_{ik} e_i = \sum_i (AP)_{ij} e_i. \quad (0.4.18)$$

Second, use the new-basis matrix and then express the new basis vectors in the old basis:

$$T(e'_j) = \sum_k A'_{kj} e'_k = \sum_k A'_{kj} \sum_i P_{ik} e_i = \sum_i (P A')_{ij} e_i. \quad (0.4.19)$$

Both are expansions of the same vector in the same basis $\{e_i\}$, so by uniqueness of coordinates the coefficients agree: $AP = P A'$, and therefore

$$\boxed{A' = P^{-1}AP} \quad (0.4.20)$$

Two matrices related this way are called **similar**, and the content of (0.4.20) is: *similar matrices are the same map, seen twice*. (Sanity check in coordinates: if $\mathbf{y} = A\mathbf{x}$ then $\mathbf{y}' = P^{-1}\mathbf{y} = P^{-1}AP\mathbf{x}'$ — the sandwich reads right to left as "translate the new coordinates into old, apply the map, translate back".)

Everything that is genuinely *about the map* must be unchanged by (0.4.20), since P encodes only our change of mind. §5 and §6 will prove that the determinant and the trace are two such invariants; Chapter 0.5 adds the eigenvalues and the characteristic polynomial. Conversely, anything that changes under (0.4.20) — any individual entry, for instance — is a fact about your description, not about the physics.

△ WHY THIS ISN'T OBVIOUS

(a) The matrix is not the map. A single linear map has *infinitely many* matrices — one for every basis, and there are infinitely many bases. Consider reflection of the plane in the line $y = x$. In the standard basis its matrix is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which looks like a swap. In the basis $e'_1 = (1, 1)$, $e'_2 = (1, -1)$ — along the mirror and perpendicular to it — the matrix is $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, which looks like "keep one direction, flip the other", and is a much better description of what the map actually does. You can check with $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ that these are related by (0.4.20). Same map. Different arrays. Note that the trace is 0 and the determinant -1 in both.

Confusing the two is the single most common failure mode in everything downstream. It is why people think a tensor "is" an array of numbers (Chapter 2.4), why the Schrödinger and Heisenberg pictures look like different physics instead of different bases (Chapter 4.4), and why gauge transformations look like magic instead of like changes of description (Chapter 6.3).

(b) Matrices do not commute, and the reason is not algebraic. Take $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, rotation by 90° , and $F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, reflection in the x -axis. Then

$$FR = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \quad RF = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad [R, F] \equiv RF - FR = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} \neq 0.$$

These are genuinely different operations: FR is a reflection in the line $y = -x$, RF a reflection in the line $y = +x$. Follow the single vector $e_1 = (1, 0)$: rotating first sends it to $(0, 1)$ and then reflecting sends it to $(0, -1)$; reflecting first leaves it at $(1, 0)$ and then rotating sends it to $(0, +1)$. Same start, opposite finish. Pick up a book and try it: turn it 90° then flip it, versus flip then turn. **Matrix multiplication fails to commute because composing transformations fails to commute.** The algebra is only reporting.

The quantity $[A, B] = AB - BA$ measuring the failure is the **commutator**, and its non-vanishing is not a nuisance to be tidied away. It is most of modern physics. $[\hat{x}, \hat{p}] = i\hbar$ is the uncertainty principle (Chapter 4.6) — measuring position then momentum is a different operation from momentum then position, by an amount $\hbar$. The commutator of two covariant derivatives is the curvature of spacetime (Chapter 3.4) — transporting a vector east-then-north differs from north-then-east, and the difference is gravity. And the commutator of two gauge fields is why gluons interact with each other while photons do not (Chapter 6.4), which is why quarks are confined and you are made of protons rather than a plasma. Every one of those is this box.

5 · The determinant, as signed volume

The determinant is usually introduced as a formula involving alternating signs and minors, from which nothing at all is obvious. We will do it the other way: state the three properties that characterise it, notice that they are the properties of *volume*, and get the formulas as a consequence.

Think of an $n \times n$ matrix A as an ordered list of its columns $a_1, \dots, a_n$, each a vector in $\mathbb{R}^n$. The three properties are:

$$\begin{aligned}
 \text{(D1) Multilinear: } & D(\dots, \alpha a_j + \beta a'_j, \dots) \\
 & = \alpha D(\dots, a_j, \dots) + \beta D(\dots, a'_j, \dots) \\
 \text{(D2) Alternating: } & \text{swapping any two columns flips the sign of } D \\
 \text{(D3) Normalised: } & D(e_1, e_2, \dots, e_n) = 1
 \end{aligned} \tag{0.4.21}$$

(D1) says: linear in each column separately, with the others held fixed. Note immediately a consequence of (D2): if two columns are *equal*, swapping them changes nothing, yet must flip the sign, so $D = -D$ and therefore $D = 0$.

Repeated column $\Rightarrow$ zero.

These are the properties of volume

Let $\text{Vol}(a_1, \dots, a_n)$ be the *signed* volume of the parallelepiped spanned by the columns from a common corner — in two dimensions, the area of the parallelogram with edges a_1, a_2 ; in three, the volume of the skewed box. We take from geometry exactly one fact: *volume equals base times perpendicular height*, the fact that makes a leaning stack of paper occupy the same volume as a straight one. Then:

(D3) is the choice of unit. The unit cube, spanned by $e_1, \dots, e_n$, has volume 1. That is what "unit" means.

(D2) is orientation. The parallelepiped spanned by (a_1, a_2) and the one spanned by (a_2, a_1) are the same set of points, but the ordered lists have opposite handedness — going from the first edge to the second is counterclockwise in one case and clockwise in the other. Signed volume records that with a sign. Consistently, if two edges coincide the solid is flat and its volume is genuinely zero, matching the consequence we just derived from (D2).

(D1) is base times height. Hold all columns but a_j fixed. They span a "base" — an $(n - 1)$ -dimensional parallelepiped of some fixed content B lying in a hyperplane through the origin. The volume of the whole solid is $B \times h$, where h is the signed perpendicular height of a_j above that hyperplane. But h is the component of a_j along the fixed unit normal $\hat{n}$ of the base, and taking a component is a linear operation: $h(\alpha a_j + \beta a'_j) = \alpha h(a_j) + \beta h(a'_j)$. Multiply by the constant B and you have exactly (D1). (If the base is itself degenerate, $B = 0$ and both sides vanish.)

So signed volume satisfies (D1)–(D3). Now the key structural fact, proved in the grind box: those three properties determine the function *completely* — there is exactly one D satisfying them. Since $\det$ and signed volume both satisfy them, they are the same thing:

$$\det A = \text{the signed volume of the parallelepiped spanned by the columns of } A.$$

(0.4.22)

And since (0.4.9) makes the columns of A the images Ae_j of the standard basis vectors, that parallelepiped is the image of the unit cube. So the sentence to carry away is: **$\det A$ is the factor by which the map A multiplies volumes, with a sign recording whether it preserves or reverses orientation.** The interactive below lets you drive that with your hands.

▼ Grind box — uniqueness, the explicit formula, and the 2×2 and 3×3 cases

Uniqueness. Suppose D satisfies (D1)–(D3). Expand every column in the standard basis, $a_j = \sum_i A_{ij}e_i$, and use multilinearity in each of the n slots in turn. Every slot contributes a sum over which basis vector was chosen there, so

$$D(a_1, \dots, a_n) = \sum_{i_1=1}^n \cdots \sum_{i_n=1}^n A_{i_1 1} A_{i_2 2} \cdots A_{i_n n} D(e_{i_1}, \dots, e_{i_n}).$$

Any term in which two of the i 's coincide has a repeated column and dies. So only the terms where $(i_1, \dots, i_n)$ is a *permutation* σ of $(1, \dots, n)$ survive. Each such $D(e_{\sigma(1)}, \dots, e_{\sigma(n)})$ can be sorted back into $D(e_1, \dots, e_n) = 1$ by a sequence of swaps, each costing a factor -1 by (D2). Writing $\text{sgn}(\sigma) = (-1)^{\#\text{swaps}}$,

$$\det A = \sum_{\sigma} \text{sgn}(\sigma) A_{\sigma(1)1} A_{\sigma(2)2} \cdots A_{\sigma(n)n},$$

which is *determined* — no freedom was left anywhere. This is the Leibniz formula, and the derivation is simultaneously the proof that at most one such D exists.

Is sgn well defined? It had better not depend on which sequence of swaps you used. Count the *inversions* of σ : the pairs $i < j$ with $\sigma(i) > \sigma(j)$. Swapping two adjacent entries changes that count by exactly ± 1 ; a swap of two entries d apart can be built from $2d - 1$ adjacent swaps, so any single swap changes the inversion count by an odd number. Hence the parity of the number of swaps needed is fixed by σ alone, and $\text{sgn}(\sigma) = (-1)^{\#\text{inversions}}$.

Existence. Uniqueness is worthless if nothing satisfies the axioms, so check that the Leibniz formula does. Each term contains exactly one factor from each column, so it is linear in each column: (D1). ✓ Swapping columns p, q replaces each σ by σ composed with a transposition, which is a relabelling of the sum that flips every sign: (D2). ✓ For $A = I$ every factor $\delta_{\sigma(j)j}$ forces $\sigma = \text{id}$, leaving the single term 1: (D3). ✓

Rows or columns? Substituting $i = \sigma(j)$ in the product shows $\det A^{\top} = \sum_{\sigma} \text{sgn}(\sigma) \prod_i A_{\sigma^{-1}(i)i}$; and σ^{-1} has the same parity as σ (reverse the order of the same swaps), while $\sigma \mapsto \sigma^{-1}$ merely reshuffles which terms are summed. So $\det A^{\top} = \det A$: every statement about columns is equally a statement about rows.

The 2×2 case, straight from the axioms. No formula needed — just expand bilinearly and kill the repeats:

$$D(ae_1 + ce_2, be_1 + de_2) = \underbrace{abD(e_1, e_1)}_0 + \underbrace{adD(e_1, e_2)}_1 + \underbrace{cbD(e_2, e_1)}_{-1} + \underbrace{cdD(e_2, e_2)}_0 = ad - bc.$$

The 3×3 case. Six permutations of $(1, 2, 3)$: three even $(123, 231, 312)$ and three odd $(213, 132, 321)$. The Leibniz formula gives

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = aei + bfg + cdh - ceg - bdi - afh,$$

which regroups into the cofactor expansion you may have memorised, $a(ei - fh) - b(di - fg) + c(dh - eg)$. Both are the same six products. The alternating signs of the cofactor rule are (D2) in disguise, and the reason the formula has $n!$ terms is that it is a sum over permutations — which is also why nobody computes a large determinant this way.

Triangular matrices. If $A_{ij} = 0$ whenever $i < j$ (lower triangular), then any $\sigma \neq \text{id}$ must somewhere send a column j to a row $i < j$, contributing a zero factor. Only the identity permutation survives: $\det A = A_{11}A_{22} \cdots A_{nn}$. That is the fact the Familiar Ground box used to read off the eigenvalues of the transition matrix.

The product rule, in one line

Now the payoff for having done it geometrically. What is $\det(AB)$? The map B multiplies every volume by $\det B$; the map A then multiplies whatever it is given by $\det A$; doing both multiplies volumes by the product. Hence

$$\boxed{\det(AB) = \det A \det B} \tag{0.4.23}$$

That is the entire proof, and it is worth pausing on how much better it is than the algebraic version. Nothing is being manipulated; the identity is a statement that *scaling factors multiply*, which is a fact about the word "factor". (The airtight version, for readers who want the geometric hand-waving cashed out into axioms, is in the next grind box — it is four lines.)

Two immediate consequences. First, taking $B = A^{-1}$: $\det(A) \det(A^{-1}) = \det I = 1$, so $\det(A^{-1}) = 1 / \det A$ — undoing a map undoes its volume scaling, as it must. Second, the determinant is **basis independent**:

$$\det(P^{-1}AP) = \det(P^{-1}) \det A \det P = \frac{1}{\det P} \det A \det P = \det A. \tag{0.4.24}$$

So the determinant is not a property of the array; it is a property of the *map*, which is why it will still make sense in Chapter 3.3 where there is no preferred basis anywhere.

▼ Grind box — the product rule without the hand-waving

Fix A and define $D(B) \equiv \det(AB)$, regarded as a function of the columns $b_1, \dots, b_n$ of B . Column j of AB is Ab_j , by (0.4.10). Now check the three axioms:

(D1) $b_j \mapsto Ab_j$ is linear, and $\det$ is linear in each column, so the composite is linear in each b_j . ✓

(D2) Swapping b_p and b_q swaps columns Ab_p and Ab_q of AB , which flips the sign of $\det(AB)$. ✓

(D3) Not normalised: $D(I) = \det(A)$, not 1. Fix that by dividing. If $\det A \neq 0$, the function $\tilde{D}(B) = D(B)/\det A$ satisfies all three axioms, so by the uniqueness proved above $\tilde{D} = \det$, i.e. $\det(AB) = \det A \det B$. ✓

If $\det A = 0$ the division is illegal, but then the columns of A are dependent (proved just below), so the columns of AB — which are combinations of the columns of A — are dependent too, and both sides are 0. The identity holds in every case. ■

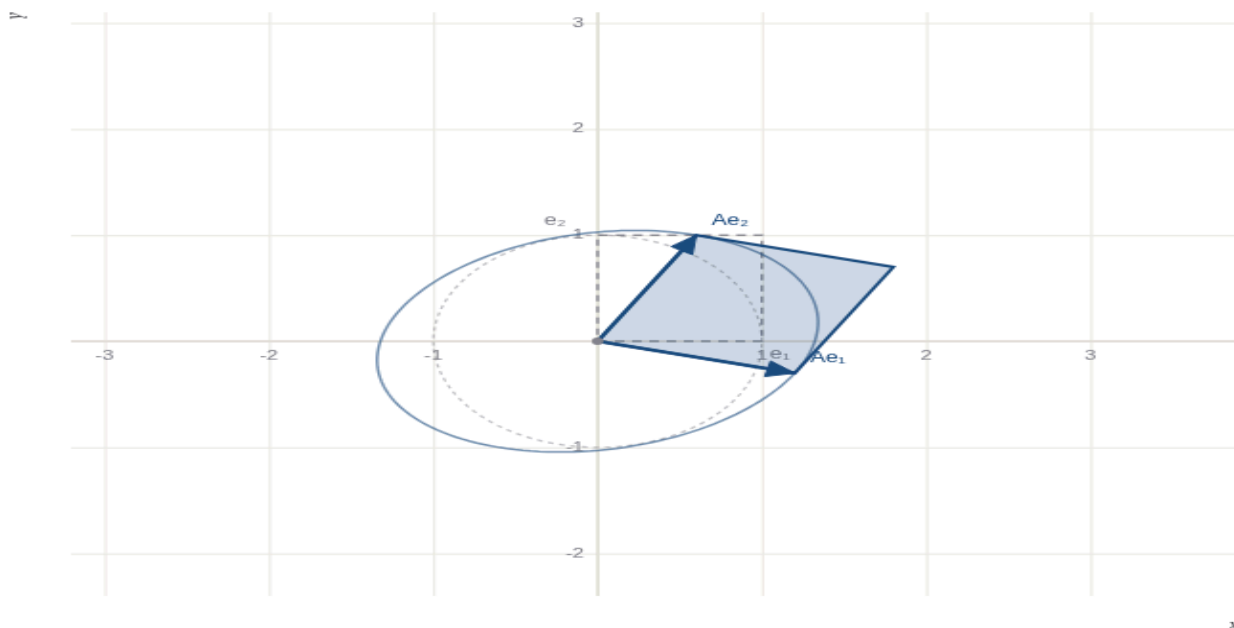
Zero determinant means information destroyed

The single most-used fact about determinants:

$$\det A \neq 0 \iff A \text{ is invertible.} \quad (0.4.25)$$

Both directions are now cheap. ($\Leftarrow$) If A is invertible then $\det A \det(A^{-1}) = 1$, so $\det A \neq 0$. ($\Rightarrow$) Contrapositive: suppose A is not invertible. Then its columns are not independent — if they were, the corollary in the §2 grind box says n independent vectors in $\mathbb{R}^n$ form a basis, so every vector in $\mathbb{R}^n$ is $A\mathbf{x}$ for exactly one $\mathbf{x}$, and the map $\mathbf{y} \mapsto \mathbf{x}$ that undoes it is linear and is an inverse. So some column is a combination of the others, say $a_1 = \sum_{j \geq 2} c_j a_j$. Expanding by multilinearity in the first slot gives a sum of determinants, each with a repeated column, each therefore zero. Hence $\det A = 0$. ■

The geometry says the same thing more vividly. $\det A = 0$ means the image of the unit cube has *no volume*: the map has flattened n -dimensional space onto something lower-dimensional — a plane, a line, a point. A whole direction's worth of information has been thrown away, and no map can retrieve it, because many different inputs now share an output. **Non-invertible is not a technicality about matrices. It is destruction of information, visible as the collapse of a volume.** That is exactly what the "singular" preset in the figure below does, and you can watch it happen by hand.



a

b

c

d

det = 1.380 (area = 1.380) tr = 2.20

presets

rotation

shear

reflection

singular

A 2×2 matrix, doing its job. The dashed grey square and circle are the originals; the solid shapes are their images under $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The two arrows are the *columns* of A — that is, the images Ae_1 and Ae_2 of the basis vectors, which is all a matrix ever is — see (0.4.9). The shaded area is $|\det A|$: the area of the original square — namely 1 — times the factor the map applies to every area, which is the whole content of (0.4.22). **Three things to do.** (i) Hit *rotation*: the square is rigidly turned, area unchanged, $\det = 1$. (ii) Hit *reflection*, and watch the shape flip through the axis while $\det$ goes negative — the sign is orientation, nothing else. (iii) Hit *singular*, or just tune the sliders until $ad = bc$: the parallelogram collapses onto a line, the circle collapses with it, the area goes to zero, and two-dimensional space has been crushed into one. Everything that used to lie on a line perpendicular to that image now lands on a single point, so the map cannot be undone. That is (0.4.25), made visible. The trace readout is there for §6; note that it is *not* the area, and that shearing changes the trace while leaving the area alone.

6 · Trace, and the identity that matters

The **trace** of a square matrix is the sum of its diagonal entries:

$$\operatorname{tr} A \equiv \sum_{i=1}^n A_{ii}. \quad (0.4.26)$$

This looks like the least interesting definition in the chapter. It is the most useful one in the book. Everything follows from a single identity.

The cyclic property

Write out both sides using (0.4.13) and swap the order of two finite sums:

$$\begin{aligned}\operatorname{tr}(AB) &= \sum_i (AB)_{ii} = \sum_i \sum_j A_{ij} B_{ji} \\ &= \sum_j \sum_i B_{ji} A_{ij} = \sum_j (BA)_{jj} = \operatorname{tr}(BA).\end{aligned}\tag{0.4.27}$$

The only thing used was that scalars commute — $A_{ij}B_{ji} = B_{ji}A_{ij}$ — even though the *matrices* emphatically do not. This is why the identity is startling: $AB \neq BA$ in general, yet their traces always agree. (Worked example 1 shows a pair where the two products differ in every entry and the traces are equal anyway.)

An immediate consequence, and the reason trace is worth anything at all:

$$\operatorname{tr}(P^{-1}AP) = \operatorname{tr}((AP)P^{-1}) = \operatorname{tr}(A),\tag{0.4.28}$$

taking $X = P^{-1}$ and $Y = AP$ in (0.4.27). **The trace is basis independent.** Like the determinant, it is a property of the map, not of the array — even though its definition mentions only diagonal entries, and "diagonal" is meaningless without a basis.

Trace is the infinitesimal determinant

Now the identity this section exists for. Take any A and any small number ϵ , and ask for the determinant of the matrix that is *almost* the identity:

$$\boxed{\det(I + \epsilon A) = 1 + \epsilon \operatorname{tr} A + O(\epsilon^2)}\tag{0.4.29}$$

Here is the honest derivation for general n , which needs nothing but the Leibniz formula from the \$5 grind box. Write $M = I + \epsilon A$, so $M_{ij} = \delta_{ij} + \epsilon A_{ij}$, and

$$\det M = \sum_{\sigma} \operatorname{sgn}(\sigma) \prod_{i=1}^n (\delta_{\sigma(i)i} + \epsilon A_{\sigma(i)i}).\tag{0.4.30}$$

Split the sum into the identity permutation and everything else.

Every non-identity permutation contributes only at order ϵ^2 . If $\sigma \neq \operatorname{id}$ then it moves at least one index — and therefore at least *two*, since a permutation cannot displace exactly one thing and leave everything else fixed (if $\sigma(i) \neq i$ then whatever was at $\sigma(i)$ has been displaced as well). For each moved index i the Kronecker delta $\delta_{\sigma(i)i}$

vanishes and the factor is $\epsilon A_{\sigma(i)i}$, carrying one power of ϵ . Two or more moved indices means two or more powers of ϵ . So all of these terms are $O(\epsilon^2)$ and none of them can affect the first-order answer.

The identity permutation gives the trace. With $\sigma = \text{id}$ every delta is 1 and $\text{sgn} = +1$, leaving

$$\prod_{i=1}^n (1 + \epsilon A_{ii}) = 1 + \epsilon \sum_{i=1}^n A_{ii} + O(\epsilon^2), \quad (0.4.31)$$

because expanding the product means choosing, from each of the n factors, either the 1 or the ϵA_{ii} : taking all the 1s gives the leading 1; taking ϵA_{ii} from exactly one factor and 1 from the rest gives ϵA_{ii} , and summing over which factor gives $\epsilon \text{tr} A$; any choice of two or more is $O(\epsilon^2)$. Adding the two contributions gives (0.4.29). ■

WHAT THIS IDENTITY SAYS

The trace is the infinitesimal version of the determinant. The determinant is the factor by which a map multiplies volume; the trace is the *rate* at which a map changes volume when it barely does anything at all. One is a multiplicative statement about a transformation, the other an additive statement about a tendency, and (0.4.29) is the bridge — exactly the relationship $e^x \approx 1 + x$ has to e , which is not a coincidence, as we are about to see.

Where this gets spent — four places, immediately

1. The divergence of a vector field is the trace of its Jacobian. Let a fluid have velocity field $v(x)$. In a short time ϵ each particle moves from x to $x + \epsilon v(x)$. That map has derivative matrix (Chapter 0.6 will call it the Jacobian) $I + \epsilon J$, where $J_{ij} = \frac{\partial v_i}{\partial x_j}$, and Chapter 0.6 will also show that the determinant of a map's derivative matrix is precisely its local volume factor — (0.4.22) applied to the linear approximation. So the volume of a small blob is multiplied, in time ϵ , by

$$\begin{aligned} \det(I + \epsilon J) &= 1 + \epsilon \text{tr} J + O(\epsilon^2) \\ &= 1 + \epsilon \sum_i \frac{\partial v_i}{\partial x_i} + O(\epsilon^2) = 1 + \epsilon (\nabla \cdot v) + O(\epsilon^2). \end{aligned} \quad (0.4.32)$$

So the divergence — which arrives in Chapter 0.7 looking like an arbitrary sum of partial derivatives — *is the fractional rate of change of volume*, and it is that because trace is the infinitesimal determinant. "Incompressible" and "divergence-free" are the same sentence.

2. Liouville's theorem is this identity. In Chapter 1.3 the state of a mechanical system is a point in phase space (q, p) , moving with velocity $\dot{q} = \frac{\partial H}{\partial p}$, $\dot{p} = -\frac{\partial H}{\partial q}$. That is a vector field on phase space, and its divergence is

$$\nabla \cdot v = \frac{\partial}{\partial q} \left(\frac{\partial H}{\partial p} \right) + \frac{\partial}{\partial p} \left(-\frac{\partial H}{\partial q} \right) = \frac{\partial^2 H}{\partial q \partial p} - \frac{\partial^2 H}{\partial p \partial q} = 0, \quad (0.4.33)$$

because mixed partial derivatives commute (Chapter 0.6). The trace of the Jacobian vanishes identically, so by (0.4.32) phase-space volume is exactly conserved by Hamiltonian flow. That is Liouville's theorem, one of the deepest facts in classical mechanics, and it is three lines of §6.

3. $\det e^A = e^{\text{tr}A}$. Define the matrix exponential by the series $e^{tA} = \sum_k t^k A^k / k!$, which converges for every t (Chapter 0.3 — the terms are bounded by those of a convergent scalar exponential series). Differentiating term by term, legitimate for a power series in t , gives $\frac{d}{dt} e^{tA} = A e^{tA}$, so to first order in ϵ

$$e^{(t+\epsilon)A} = e^{tA} + \epsilon A e^{tA} + O(\epsilon^2) = (I + \epsilon A + O(\epsilon^2)) e^{tA}. \quad (0.4.34)$$

Let $f(t) \equiv \det e^{tA}$. Take determinants of (0.4.34) and use the product rule (0.4.23) together with (0.4.29) — legitimate because $\det$ is a polynomial in the entries, so an $O(\epsilon^2)$ change in the entries makes an $O(\epsilon^2)$ change in the determinant:

$$\begin{aligned} f(t + \epsilon) &= (1 + \epsilon \text{tr}A + O(\epsilon^2)) f(t) \\ \implies f'(t) &= (\text{tr}A) f(t), \quad f(0) = \det I = 1. \end{aligned} \quad (0.4.35)$$

By Chapter 0.1, the function equal to a constant times its own derivative, starting at 1, is the exponential. Hence $f(t) = e^{t \text{tr}A}$, and at $t = 1$

$$\boxed{\det e^A = e^{\text{tr}A}} \quad (0.4.36)$$

4. Why the generators of $\text{SU}(N)$ are traceless. A continuous symmetry is built by exponentiating a generator: $U(\theta) = e^{\theta X}$. The "S" in $\text{SU}(N)$ stands for *special*, meaning $\det U = 1$ — the transformations preserve volume. By (0.4.36) that requires $e^{\theta \text{tr}X} = 1$ for all θ , i.e.

$$\det U = 1 \text{ for all } \theta \iff \text{tr} X = 0. \quad (0.4.37)$$

This is not a bookkeeping convention. In Chapter 6.1 you will count the anti-Hermitian 3×3 matrices and find a 9-dimensional space; tracelessness is one real linear condition, which removes exactly one dimension, leaving $3^2 - 1 = 8$. There are eight gluons because of (0.4.29). The most consequential appearance of a sum of diagonal entries in physics is a particle count.

7 · Why complex vector spaces

Everything so far worked equally over $\mathbb{R}$ and over $\mathbb{C}$. Here is where they part, and the difference is not aesthetic.

An **eigenvector** of an operator T is a nonzero v with $Tv = \lambda v$ — a direction the map leaves alone, stretching by the factor λ . These are the directions in which a linear map is simple, and Chapter 0.5 is devoted to finding them.

Over $\mathbb{R}$, they need not exist. Let R be rotation of the plane by 90° . If $Rv = \lambda v$ for real λ and nonzero real v , apply R again: $R^2v = \lambda^2v$. But rotating twice by 90° is $R^2 = -I$, so $\lambda^2v = -v$, giving $\lambda^2 = -1$ — with no real solution. The geometric statement is obvious once said: *a rotation leaves no direction pointing along itself*. That is what rotating means.

Over $\mathbb{C}$, they always exist. Let A be any $n \times n$ complex matrix, $n \geq 1$. Consider

$$p(\lambda) \equiv \det(A - \lambda I). \quad (0.4.38)$$

By the Leibniz formula this is a polynomial in λ of degree exactly n : the identity permutation contributes $\prod_i (A_{ii} - \lambda)$, whose leading term is $(-\lambda)^n$, and every other permutation misses at least two diagonal factors and so has degree at most $n - 2$ — nothing can cancel the leading term. By the fundamental theorem of algebra, every non-constant polynomial with complex coefficients has a complex root. (This is the one result in this chapter we import rather than prove; it is a theorem of complex analysis, and Chapter 0.3's discussion of the complex plane is the right place to have made peace with it.) So there is a $\lambda_0 \in \mathbb{C}$ with $\det(A - \lambda_0 I) = 0$, and then by (0.4.25) the matrix $A - \lambda_0 I$ is not invertible, so some nonzero v has $(A - \lambda_0 I)v = 0$, that is

$$Av = \lambda_0 v. \quad (0.4.39)$$

Every operator on a complex vector space has at least one eigenvalue. Over $\mathbb{R}$ that statement is simply false, and the 90° rotation is the counterexample.

Watch what the complex numbers do to that rotation. Over $\mathbb{C}$ the rotation by angle θ , $R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$, acting on the complex vector $(1, -i)$, gives

$$R_\theta \begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} \cos \theta + i \sin \theta \\ \sin \theta - i \cos \theta \end{pmatrix} = (\cos \theta + i \sin \theta) \begin{pmatrix} 1 \\ -i \end{pmatrix} = e^{i\theta} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \quad (0.4.40)$$

The eigenvalue of a rotation is $e^{i\theta}$ — Euler's formula from Chapter 0.3, arriving as the spectrum of a rotation matrix. The rotation that had no real eigenvector has two complex ones, with eigenvalues $e^{\pm i\theta}$ of unit modulus. Rotation and complex phase are the same phenomenon written twice, which is why Chapter 0.3 could say $U(1)$ and $SO(2)$ are the same group.

This is why quantum mechanics is built on complex vector spaces rather than real ones. Physical quantities in Chapter 4.2 are represented by operators, and their measurable values are eigenvalues; a theory in which operators can fail to have any is not a theory of measurement. And time evolution is a rotation in state space (Chapter 0.1's closing observation, now precise): $e^{-i\hat{H}t/\hbar}$ has unit-modulus eigenvalues, so it preserves lengths, so probability is conserved. Take away the complex numbers and you lose the eigenvalues, the phases, and the conservation of probability together.

8 · Worked examples

WORKED EXAMPLE 1 — MATRIX MULTIPLICATION, DERIVED ON A CONCRETE PAIR

Let T be the shear $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ and S the map $\begin{pmatrix} 3 & 0 \\ 1 & -1 \end{pmatrix}$, both from $\mathbb{R}^2$ to $\mathbb{R}^2$ in the standard basis. Find the matrix of $S \circ T$ by tracking basis vectors only, then check against (0.4.13).

First read the columns as images of basis vectors, which is all (0.4.9) says:

$$\begin{aligned} T e_1 &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} = e_1, & T e_2 &= \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 2e_1 + e_2; \\ S e_1 &= \begin{pmatrix} 3 \\ 1 \end{pmatrix}, & S e_2 &= \begin{pmatrix} 0 \\ -1 \end{pmatrix}. \end{aligned}$$

Now push the basis vectors through both maps in turn, using linearity of S and *nothing else*:

$$\begin{aligned} (S \circ T) e_1 &= S(e_1) = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \\ (S \circ T) e_2 &= S(2e_1 + e_2) = 2 S e_1 + S e_2 = 2 \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}. \end{aligned}$$

Those two columns *are* the matrix of the composite: $ST = \begin{pmatrix} 3 & 6 \\ 1 & 1 \end{pmatrix}$. Check against the formula $(ST)_{ik} = \sum_j S_{ij} T_{jk}$:

$$\begin{aligned} (ST)_{11} &= 3 \cdot 1 + 0 \cdot 0 = 3, & (ST)_{12} &= 3 \cdot 2 + 0 \cdot 1 = 6, \\ (ST)_{21} &= 1 \cdot 1 + (-1) \cdot 0 = 1, & (ST)_{22} &= 1 \cdot 2 + (-1) \cdot 1 = 1. \quad \checkmark \end{aligned}$$

Identical, as it must be — the formula was *derived* from this procedure, so the check is really a check that we can add.

Now reverse the order. $TS = \begin{pmatrix} 5 & -2 \\ 1 & -1 \end{pmatrix}$, which shares not one entry with ST . Composition does not commute, and here is the visible reason: ST means shear first, TS means shear last, and a shear applied to an already-distorted figure does something different from a shear applied to the original.

Two invariants survive anyway. Determinants: $\det S = -3$, $\det T = 1$, and $\det(ST) = 3 \cdot 1 - 6 \cdot 1 = -3 = \det(TS) = 5(-1) - (-2)(1)$. Both equal $\det S \det T$, exactly as (0.4.23) requires — volume scalings multiply and do not care about order, because numbers commute even when maps do not. Traces: $\text{tr}(ST) = 3 + 1 = 4$ and $\text{tr}(TS) = 5 - 1 = 4$. Equal, as (0.4.27) promised, despite every single entry being different. That is the cyclic identity earning its keep.

WORKED EXAMPLE 2 — $\det(I + \epsilon A)$ FOR A 3×3 , WITH THE ϵ^2 TERMS EXPOSED

Verify (0.4.29) explicitly for $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & -1 & 3 \\ 1 & 4 & 5 \end{pmatrix}$, whose trace is $2 - 1 + 5 = 6$.

Write out the perturbed matrix and expand the determinant exactly — no approximation anywhere, using the 3×3 formula from the \$5 grind box:

$$I + \epsilon A = \begin{pmatrix} 1 + 2\epsilon & \epsilon & 0 \\ 0 & 1 - \epsilon & 3\epsilon \\ \epsilon & 4\epsilon & 1 + 5\epsilon \end{pmatrix}$$

$$\begin{aligned} \det(I + \epsilon A) &= (1 + 2\epsilon) \left[(1 - \epsilon)(1 + 5\epsilon) - 12\epsilon^2 \right] - \epsilon \left[0 - 3\epsilon^2 \right] + 0 \\ &= (1 + 2\epsilon)(1 + 4\epsilon - 17\epsilon^2) + 3\epsilon^3 \\ &= 1 + 6\epsilon - 9\epsilon^2 - 31\epsilon^3. \end{aligned}$$

The coefficient of ϵ is $6 = \text{tr } A$. ✓ The identity holds, and we now have the whole polynomial rather than just its first two terms, so we can see precisely what was discarded.

Where the ϵ^2 terms come from. Two sources, exactly as the general derivation said. First, the identity permutation: taking ϵA_{ii} from two of the three diagonal factors in (0.4.31) gives $\epsilon^2(A_{11}A_{22} + A_{11}A_{33} + A_{22}A_{33}) = \epsilon^2(-2 + 10 - 5) = 3\epsilon^2$. Second, the transpositions — permutations that swap exactly two indices and fix the rest. The swap $1 \leftrightarrow 3$ contributes $-\epsilon^2 A_{13}A_{31}(1 + \epsilon A_{22})$, and so on; at order ϵ^2 these give $-\epsilon^2(A_{12}A_{21} + A_{13}A_{31} + A_{23}A_{32}) = -\epsilon^2(0 + 0 + 12) = -12\epsilon^2$. Total: $3 - 12 = -9$. ✓

Combining the two sources, the ϵ^2 coefficient is $\sum_{i < j} (A_{ii}A_{jj} - A_{ij}A_{ji})$ — the sum of the principal 2×2 minors, $-2 + 10 - 17 = -9$. ✓ And the ϵ^3 coefficient is $\det A = -31$, which it must be, since taking ϵA from every slot is the same as scaling A by ϵ in all three columns. The pattern continues for every n : the coefficient of ϵ^k in $\det(I + \epsilon A)$ is the sum of the principal $k \times k$ minors, starting at 1, then $\text{tr } A$, and ending at $\det A$. **Trace and determinant are the first and last members of one family**, which is the family Chapter 0.5 will identify with the coefficients of the characteristic polynomial.

How good is first order? At $\epsilon = 0.01$ the exact value is 1.059069 against the linear estimate $1 + 6\epsilon = 1.06$: an error of -9.31×10^{-4} , where the quadratic term alone predicts $-9\epsilon^2 = -9.00 \times 10^{-4}$. Agreement to two figures, with the small residue supplied by the cubic term — exactly the behaviour the $O(\epsilon^2)$ notation promises. This is Chapter 0.1's linearization applied to a function of nine variables, and it behaves exactly as advertised.

9 · Your turn

PROBLEM 1 — A TWO-DIMENSIONAL SPACE WITH NO ARROWS IN IT

Show that the set of solutions of $y'' + \omega^2 y = 0$ (with $\omega \neq 0$ fixed) is a vector space, that $\{\cos \omega t, \sin \omega t\}$ is a basis for it, and hence that its dimension is 2. Then say what the "two arbitrary constants" of your first differential-equations course actually are.

– Solution

Vector space. This is (0.4.3) with $p = 0, q = \omega^2$: if y_1, y_2 are solutions then $(ay_1 + by_2)'' + \omega^2(ay_1 + by_2) = a(y_1'' + \omega^2 y_1) + b(y_2'' + \omega^2 y_2) = 0$. The zero function is a solution, so the set is nonempty and closed under both operations. ✓

Both candidates are solutions. $\frac{d^2}{dt^2} \cos \omega t = -\omega^2 \cos \omega t$ and likewise for $\sin$ (Chapter 0.1). ✓

Independent. Suppose $a \cos \omega t + b \sin \omega t = 0$ for all t . Put $t = 0$: $a = 0$. Differentiate and put $t = 0$: $b\omega = 0$, and $\omega \neq 0$, so $b = 0$. ✓

Spanning — the only part with any content. Let y be *any* solution. Set $c_1 = y(0)$ and $c_2 = y'(0)/\omega$, and define $u = y - c_1 \cos \omega t - c_2 \sin \omega t$. Then u solves the same equation (it is a combination of solutions) and satisfies $u(0) = 0, u'(0) = 0$. Now consider

$$E(t) \equiv (u')^2 + \omega^2 u^2, \quad E' = 2u'u'' + 2\omega^2 uu' = 2u'(u'' + \omega^2 u) = 0.$$

So E is constant, and $E(0) = 0$. Both terms of E are non-negative and sum to zero, so each is zero for every t ; in particular $u \equiv 0$ and $y = c_1 \cos \omega t + c_2 \sin \omega t$. ✓ (That E is the energy of a harmonic oscillator is not an accident, and Chapter 0.8 will make a habit of this trick.)

Hence the solution set has a basis with two elements and dimension 2.

The constants. They are **coordinates**. Specifying a solution means specifying a vector in a two-dimensional space, which takes exactly two numbers — and the familiar alternative form $y = C \cos(\omega t - \varphi)$ is not a different solution set but a different *parametrisation* of the same plane: expanding gives $C \cos \varphi \cos \omega t + C \sin \varphi \sin \omega t$, so $(c_1, c_2) = (C \cos \varphi, C \sin \varphi)$, which is polar coordinates on the same two-dimensional space. Amplitude-and-phase versus sine-and-cosine is a change of description, and the physics is in neither.

PROBLEM 2 — DIFFERENTIATION AS A MATRIX, AND WHY INTEGRATION NEEDS A $+C$

Let D be the derivative operator on P_3 , the polynomials of degree ≤ 3 , with basis $\{1, x, x^2, x^3\}$. Write down its matrix. Compute $\det D$ and $\operatorname{tr} D$, and interpret both.

– Solution

Apply (0.4.9): the j -th column is the image of the j -th basis vector, written in coordinates. With $e_1 = 1$, $e_2 = x$, $e_3 = x^2$, $e_4 = x^3$:

$$D(1) = 0, \quad D(x) = 1 = e_1, \quad D(x^2) = 2x = 2e_2, \quad D(x^3) = 3x^2 = 3e_3,$$

$$\text{so} \quad D = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Determinant. The first column is entirely zero, so by multilinearity (D1) the determinant is linear in that column and $D(\dots, 0, \dots) = 0$. Hence $\det D = 0$. Equivalently the matrix is triangular with zero diagonal. By (0.4.25) the operator is *not invertible*, and §5 says what that means concretely: it destroys information. Here you can name the information exactly. D kills every constant, so 1 and $1 + 7$ and $1 - \pi$ all have the same image, and no operation can tell them apart afterwards. **That is the constant of integration.** The $+C$ is not a bookkeeping nicety; it is the statement that $\det D = 0$ and that antidifferentiation is therefore not a function but a one-parameter family.

Geometrically: the image of D is P_2 , of dimension 3, so a four-dimensional space has been flattened onto a three-dimensional one — the unit hypercube of P_3 has been crushed to zero four-volume, exactly the collapse the interactive shows in two dimensions.

Trace. $\operatorname{tr} D = 0 + 0 + 0 + 0 = 0$. Consistent with (0.4.29): $\det(I + \epsilon D) = 1$ exactly, since $I + \epsilon D$ is triangular with 1s on the diagonal, and the linear term $\epsilon \operatorname{tr} D$ is indeed zero. Nudging the identity in the direction of "differentiate a bit" changes no volume at all to first order — reassuring, since D is *nilpotent* ($D^4 = 0$: differentiate a cubic four times and nothing is left), and nilpotent operators always have zero trace.

PROBLEM 3 — HOW FAR DOES THE CYCLIC PROPERTY GO?

Prove $\text{tr}(AB) = \text{tr}(BA)$ for all square A, B of the same size. Then show $\text{tr}(ABC) = \text{tr}(BCA)$ follows immediately, but that $\text{tr}(ABC) = \text{tr}(ACB)$ is *false* in general — give an explicit 2×2 counterexample.

– Solution

Two factors. This is (0.4.27): $\text{tr}(AB) = \sum_{i,j} A_{ij}B_{ji} = \sum_{j,i} B_{ji}A_{ij} = \text{tr}(BA)$, using only commutativity of the scalars and the fact that a finite double sum may be reordered.

Three factors, cyclically. Apply the two-factor result with $X = AB$ and $Y = C$:

$$\text{tr}(ABC) = \text{tr}((AB)C) = \text{tr}(C(AB)) = \text{tr}(CAB),$$

and once more with $X = A, Y = BC$ gives $\text{tr}(ABC) = \text{tr}(BCA)$. So the three *cyclic* rotations ABC, BCA, CAB all agree.

The other three orderings need not. Take

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Then $AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, so $ABC = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $\text{tr}(ABC) = 1$. But $AC = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = 0$, so $ACB = 0$ and $\text{tr}(ACB) = 0 \neq 1$. ✓

The moral. The trace is invariant under *cyclic* permutation, not under arbitrary permutation — because cycling is the only rearrangement you can reach by repeatedly moving the last factor to the front, which is all $\text{tr}(XY) = \text{tr}(YX)$ licenses. Anyone who forgets this and "simplifies" $\text{tr}(ABC)$ to $\text{tr}(ACB)$ has assumed $BC = CB$. In Chapter 5.7 you will evaluate traces of products of gamma matrices, which anticommute rather than commute; the cyclic property is what makes those traces computable, and the absence of the non-cyclic property is what makes them interesting.

PROBLEM 4 — EVERY ENTRY CHANGES, TWO NUMBERS DO NOT

Let T have matrix $A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$ in the standard basis. Change to the basis $e'_1 = e_1 + e_2, e'_2 = 2e_1 + 3e_2$. Compute $A' = P^{-1}AP$, verify directly from (0.4.17) that A' really is the matrix of the same map, and check the determinant and trace.

– Solution

By (0.4.14) the columns of P are the new basis vectors in old coordinates, so $P = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$, with $\det P = 1$ and (using the 2×2 inverse, which you can verify by multiplying out) $P^{-1} = \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix}$. Then

$$AP = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 7 \\ 7 & 18 \end{pmatrix}, \quad A' = P^{-1}(AP) = \begin{pmatrix} -5 & -15 \\ 4 & 11 \end{pmatrix}.$$

Direct check. (0.4.17) says column 1 of A' should be the coordinates of $T(e'_1)$ in the new basis. Compute the left side: $T(e'_1) = A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$. And the right side: $-5e'_1 + 4e'_2 = -5 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 4 \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$. ✓ Same vector. Likewise $T(e'_2) = A \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 7 \\ 18 \end{pmatrix}$ and $-15e'_1 + 11e'_2 = \begin{pmatrix} -15 + 22 \\ -15 + 33 \end{pmatrix} = \begin{pmatrix} 7 \\ 18 \end{pmatrix}$. ✓

Invariants.

$$\operatorname{tr} A = 2 + 4 = 6, \quad \operatorname{tr} A' = -5 + 11 = 6; \quad \det A = 8 - 3 = 5, \quad \det A' = -55 + 60 = 5.$$

Not one of the four entries survived — 2, 1, 3, 4 became $-5, -15, 4, 11$, and no entry of A' even appears in A — yet the trace and the determinant are untouched, exactly as (0.4.24) and (0.4.28) guarantee. If you were handed only A' and asked "what does this map do to areas?", you would answer 5, the same answer as from A , because that question is about the map. If you were asked "what is the top-left entry?", you would be answering a question about the basis.

This is the entire chapter in four numbers.

THE BRICK YOU JUST LAID

You have the vector space defined by what you may do to its elements rather than by what they look like, so that functions, polynomials, matrices, solutions of differential equations and quantum states are all vectors and all obey the same theorems. You have coordinates, proved unique, together with the discipline of never confusing the vector with its coordinate list. You have matrix multiplication *derived* from composition of maps — including why the shapes must match, why it is associative for free, and why it does not commute. You have the change-of-basis law $A' = P^{-1}AP$ and the distinction between what changes with the description and what does not. You have the determinant as signed volume, from three properties rather than a formula, with $\det(AB) = \det A \det B$ as a one-line statement that scaling factors multiply, and $\det = 0$ as the destruction of information. And you have the trace: cyclic, basis independent, and — the identity worth the whole chapter — the first-order determinant, $\det(I + \epsilon A) = 1 + \epsilon \operatorname{tr} A + O(\epsilon^2)$.

Where this gets spent. *Basis and coordinates* → Chapter 2.4 (a tensor is defined by its transformation law, which is §4) and Chapter 3.2 (physics must not depend on coordinates). *Composition of linear maps* → Chapter 0.6, where the chain rule becomes a product of Jacobian matrices and is the same statement as (0.4.13). *Determinant* → Chapter 0.6 again (the Jacobian determinant in a change of variables is precisely "the factor by which volume is multiplied") and Chapter 3.3, where $\sqrt{-g}$ — the square root of a determinant — is what makes an integral over curved spacetime mean anything. *Trace* → Chapter 0.7 (divergence is the trace of the Jacobian), Chapter 1.3 (Liouville's theorem is $\operatorname{tr} = 0$), and Chapter 6.1 (the generators of $SU(N)$ are traceless because $\det e^A = e^{\operatorname{tr} A}$, which is why there are eight gluons). *Non-commutativity* → Chapter 4.6 (uncertainty), Chapter 3.4 (curvature), Chapter 6.4 (Yang–Mills self-interaction). *Complex scalars* → Chapter 4.2, where the existence of eigenvalues is the existence of measurable values.

Next, Chapter 0.5 asks the obvious follow-up question: given an operator, can we *find* a basis in which its matrix is as simple as possible? For a large and physically crucial class of operators the answer is yes — diagonal — and the diagonal entries are the eigenvalues that this chapter has twice promised.